

GEHAD FATEHI ALSHABI

CAD/CAM & Mechatronics Engineer | Full-Stack Developer | DevOps & AI Automation

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PROFESSIONAL SUMMARY

CAD/CAM and Mechatronics Engineer with hands-on experience in sheet metal design and CNC fiber laser cutting of stainless steel using SolidWorks, AutoCAD and CypCut/CypNest. Skilled in converting engineering drawings into production-ready cutting files, optimizing nesting to reduce material waste, and setting laser cutting parameters. Strong background in industrial automation, electrical control panels and solar power systems, with HSE training and a focus on quality and safe production. Also builds and operates full-stack software: a WhatsApp customer service chatbot for the company and two React Native mobile products backed by TypeScript APIs, Redis and Docker containers on a self-managed Linux server, including LLM-based automation.

SKILLS

- **CAD/CAM:** SolidWorks (3D modeling, sheet metal, flat patterns), AutoCAD (2D drafting, DXF/DWG), CypCut, CypNest
- **Manufacturing:** CNC fiber laser cutting, nesting optimization, laser cutting parameters (power, speed, focus, assist gas), stainless steel fabrication, sheet metal design for bending, shop drawings
- **Automation & Control:** Industrial control panel design and installation, classic control circuits, industrial communication protocols (Modbus, Profibus, EtherNet/IP); PLC programming (Siemens, Allen-Bradley, Schneider Electric) from university coursework and graduation project
- **Electrical & Energy:** Solar PV system design and installation, power distribution layouts, electrical troubleshooting and maintenance
- **Software Development:** React Native, Expo, TypeScript, Node.js, Express, Redis, Zod, REST APIs, WhatsApp Cloud API (Meta), Firebase Cloud Messaging, Sentry, i18n with right-to-left support
- **DevOps & Infrastructure:** Ubuntu Linux server administration, Docker and Docker Compose, Caddy reverse proxy with automatic HTTPS (Let's Encrypt), UFW/iptables firewalls, SSH hardening, Cloudflare DNS, cron scheduling, backups and monitoring, Contabo VPS
- **AI & Automation:** LLM integration in production (DeepSeek, OpenAI-compatible APIs), agentic AI coding tools (Claude Code, MCP servers, custom skills), prompt design, automated content pipelines, programmatic SVG image generation
- **Programming:** JavaScript, TypeScript, Python, C, C++, Flutter
- **Software:** MATLAB, Proteus, Multisim, Cisco Packet Tracer, VS Code, Microsoft Office (Word, Excel, PowerPoint)
- **Soft Skills:** Problem solving, communication and teamwork, time management, adaptability

PROFESSIONAL EXPERIENCE

CAD/CAM Engineer

Dec 2025 – Present

Masdar Al Maaden | Riyadh, Saudi Arabia

- Design stainless steel parts and products in SolidWorks and AutoCAD, including 3D models, sheet metal flat patterns for bending, and shop drawings.
- Convert customer and engineering drawings into production-ready 2D DXF/DWG cutting files for CNC fiber laser cutting machines.
- Nest parts on stainless steel sheets using CypNest to maximize material utilization and reduce scrap.
- Program and run cutting jobs in CypCut, setting laser power, cutting speed, focus position and assist gas for each sheet thickness.
- Coordinate with machine operators and the production team to deliver orders accurately and on schedule.
- Developed and deployed the company's WhatsApp customer service chatbot using Node.js and the WhatsApp Cloud API (Meta).
- Automated order and complaint handling by routing orders to sales management and complaints to management, and gave customers instant access to branch and workshop locations nationwide, branch ratings and company contacts.

Mechatronics Engineer

Jan 2025 – Aug 2025

Mashaal Electrical Engineering | Sana'a, Yemen

- Designed power distribution layouts for residential and commercial solar PV systems.
- Designed and installed classic control systems and control panels for industrial machines.
- Coordinated with technicians on site during mounting, wiring, installation and maintenance.
- Performed safety inspections and installation quality assessments to ensure compliance and reliable system performance.

Mechanical Engineering Trainee

United Engineering & Automobile Co. Ltd. | Sana'a, Yemen

Sep 2024 – Dec 2024

- Applied mechanical engineering principles to vehicle maintenance and repair.
- Performed technical troubleshooting and diagnosis of mechanical faults.

Electrical Technician

Solar Care for Solar Energy | Sana'a, Yemen

May 2019 – Dec 2021

- Installed and commissioned solar power systems for customers.
- Installed and maintained electrical systems and control panels.
- Performed maintenance and troubleshooting of electrical equipment.

PROJECTS

Golazo! – Live Football Scores App

Freelance project – live on Google Play

Google Play: <https://play.google.com/store/apps/details?id=com.golazo.wc2026>

- Built and published an Android app for live football scores, match data and league standings, and maintained it across seven releases, owning app signing, versioning, store listing, privacy policy and data-safety compliance.
- Developed the product as three services: a TypeScript/Express data API with a live match scanner, a content orchestrator that renders and publishes cards, and the React Native (Expo) app.
- Migrated the live product from a usage-priced platform to a flat-cost self-managed VPS with no downtime, running both systems in parallel with a tested DNS rollback plan.

Trading Index – Economic Calendar & Market Analysis App

Freelance project – release-ready

Google Play: link will be provided after release.

- Built a crypto-first economic calendar and market-impact app with push notifications (Firebase Cloud Messaging and Expo), crash reporting (Sentry) and a trilingual interface in English, Arabic and Spanish, including full right-to-left layout.
- Designed caching and scheduling around third-party API quotas: per-source Redis TTLs, thirteen deduplicated cron jobs and serve-stale fallbacks that keep endpoints healthy when an upstream source fails.
- Integrated a DeepSeek LLM into the production pipeline for summarization, classification and translation with per-item caching and fail-open behaviour, and built an SVG rendering service generating branded cards from live data across twelve templates.

Self-Managed Production Server & DevOps

Personal infrastructure for both apps

- Provisioned and hardened an Ubuntu 24.04 VPS hosting both products in Docker containers: key-only SSH, layered UFW and iptables firewalls, a shared Caddy reverse proxy with automatic Let's Encrypt HTTPS, Cloudflare DNS and email routing, nightly offsite backups and scripted deploys.
- Used agentic AI tooling (Claude Code with MCP servers, custom skills and per-repository memory files) to develop and document a multi-repository product across six repositories.

Remote I/O Module (RIOM) and Protocol Converter – Graduation Project

2024

- Designed and developed a remote I/O module with a protocol conversion system supporting Modbus, Profibus and EtherNet/IP to integrate industrial automation systems and enable remote monitoring and control.

EDUCATION

Bachelor of Science in Mechatronics Engineering

Sana'a University, Sana'a, Yemen

Jun 2024

CERTIFICATIONS & TRAINING

- HSE Engineering Specialization (2025)
- Workplace Safety and Health Diploma (2025)
- Project Management Course – PBT (2025)
- English Language Diploma – PBT (2025)
- International Computer Driving Licence (ICDL) – Sana'a University (2024)

LANGUAGES

- **Arabic:** Native
- **English:** Professional working proficiency (C1)